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SUPERSEDING
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June 21, 1956

FEDERAL SPECIFICATION

JUNCTION BOX AND COVER, JUNCTION BOX (NONMETALLIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nonmetallic junction boxes and junction-box covers (see 6.3.1) for use with open wiring on insulators, concealed knob-and-tube wiring, nonmetallic sheathed cable, and flexible nonmetallic tubing. It does not cover electrical wall plates or metallic junction boxes for use with rigid steel conduit, electrical metallic-tubing, armored cable, and flexible metallic conduit.

1.2 Classification.

1.2.1 *Junction boxes.*

1.2.1.1 *Types, classes, and styles.* Junction boxes furnished under this specification shall be of the following types, classes, and styles specified (see 6.2):

Types:

Type I—Round box.

Type II—Square box.

Type III—Rectangular box.

Type IV—Octagonal box.

Classes:

Class 1—Deep box with mounting support brackets.

Class 2—Deep box without mounting support brackets.

Class 3—Shallow box with mounting support brackets.

Class 4—Shallow box without mounting support brackets.

Styles:

Style A—Plain.

Style B—With facilities for mounting wiring devices (see 6.1).

Style C—With facilities for clamping nonmetallic cables or tubing.

Style D—With facilities for mounting wiring devices and for clamping nonmetallic cables or tubing.

1.2.2 *Junction-box covers.*

1.2.2.1 *Types and styles.* Junction-box covers (see 6.3) furnished under this specification shall be of the following types and styles as specified (see 6.2):

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Types:

Type V—Round cover.
Type VI—Square cover.
Type VII—Rectangular cover.
Type VIII—Octagonal cover.

Styles:

Style E—Blank.
Style F—Single receptacle.
Style G—Duplex receptacle.
Style H—Double duplex receptacle.
Style I—Toggle switch.
Style J—Double toggle switch.
Style K—Toggle switch and duplex receptacle.
Style L—Pendant.
Style M—Plaster ring.
Style N—Surface mounting.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

Underwriters' Laboratories, Inc. Standard:

UL–514—Standard for Outlet Boxes and Fittings.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 E Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blv., Santa Clara, California, 95050).

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3. REQUIREMENTS

3.1 Material. Junction boxes and junction-box covers furnished under this specification shall be of cold-molded composition, phenolic, or porcelain, as specified (see 6.2), and shall conform to the requirements of UL Standard No. 514.

3.2 Fire and casualty hazards.

3.2.1 Each contractor shall submit to the contracting agency proof that the junction boxes and covers he proposes to supply under this specification conform to the requirements of UL Standard No. 514. The label, or listing with reexamination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the junction boxes and covers conform to this requirement.

3.2.2 In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency that his junction boxes and covers conform to the applicable requirements of the published standards, including methods of test, of UL Standard No. 514.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Dimensions. Junction boxes shall be of sufficient dimensions to enclose devices mounted therein or thereon with ample space for wiring and fittings (see 6.4). Specific dimensions of boxes and covers shall be as specified (see 6.2).

3.4 Knockouts. Knockouts provided in the walls and backs of boxes for the entrance of tubing, cable, or wires shall be effectively closed to afford protection similar to that of the box walls. The knockouts shall be located on flat surfaces of boxes so that connectors may be clamped against the box wall.

3.5 Components. Components such as mounting brackets, clamps, and connectors furnished as part of a junction box shall be in complete accord with the requirements of UL Standard No. 514.

3.6 Identification marking. Junction boxes, covers, clamps, and connectors furnished as part of a junction box shall be marked in accordance with the requirements specified in UL Standard No. 514.

3.7 Workmanship. Each item shall bear the appearance of professional workmanship, shall be clean to handle, shall be free of sharp edges and rough projections, and shall be sturdy in assembly.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For the purposes of inspection, a lot shall consist of all junction boxes or junction-box covers or junction boxes and covers of the same type, class, style, size, and manufacture offered for delivery at one time. The sample unit shall be one junction box, one junction-box cover, or one junction box and cover as applicable.

4.3 Sampling. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated.

4.3.1 *Sampling for visual inspection.* Sampling of each lot shall be performed using an

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inspection level of II with an acceptable quality level of 4.0 percent defective for major defects and 10.0 percent defective for minor defects.

4.3.2 *Sampling for dimensional examination.* Sampling of each lot shall be performed using an inspection level of I with an acceptable quality level of 4.0 percent defective.

4.3.3 *Sampling for tests.* Sampling of each lot shall be performed using an inspection level of S–1 with an acceptance number of 0. AQL shall be 0.010 percent defective.

4.4 Examinations.

4.4.1 *Visual examination.* Samples selected in accordance with 4.3.1 shall be examined for defects listed below:

Major

101. Material not as specified.

102. Samples chipped, cracked, or broken.

103. Rough projections or sharp edges.

104. Component missing or incorrect.

105. Knockouts not povided or not as specified.

Minor

201. Marking missing, incorrect, or improper.

4.4.2 *Dimensional examination.* Samples selected in accordance with 4.3.2 shall be examined for compliance with dimensions specified in the contract.

4.5 Tests. Samples selected in accordance with 4.3.3 shall be tested in accordance with UL Standard No. 514. Failure to pass the specified tests shall be cause for rejection.

4.6 Inspection of preparation for delivery.

4.6.1 *Pilot-pack inspection.*

4.6.1.1 *Examination.* The pilot pack shall be examined in accordance with 4.6.3.1. Presence of one or more defects shall be cause for rejection.

4.6.1.2 *Tests.* The pilot pack shall be tested in accordance with 4.6.3.2. Failure of the pilot pack to pass the specified test shall be cause for rejection.

4.6.2 *Quality conformance inspection.*

4.6.2.1 *Unit of product.* For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 *Examinations.* Samples shall be selected in accordance with MIL–STD–105, and examined as specified in 4.6.3.1. The inspection level shall be S–2 with an acceptable quality level of 2.5 percent defective.

4.6.3 *Inspection procedure.*

4.6.3.1 *Examination.* The preservation, packaging, packing, and marking shall be examined for the following defects:

106. Materials not as specified for levels A and B.

107. Intermediate packaging not as specified for levels A and B.

108. Strapping not zinc coated for level A.

109. Fiberboard boxes not sealed and reinforced as specified for level B.

110. Marking illegible, incomplete, incorrect for levels A, B, and C.

4.6.3.2 *Tests.* The following tests applicable to the specified level shall be performed. At the conclusion of the pilot-pack test, the pack shall be examined. Any visible damage to the junction boxes and covers shall constitute failure of the pilot pack and shall be cause for rejection.

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4.6.3.2.1 *Level A.*

4.6.3.2.1.1 *Free-fall drop test.* The pilot pack shall be dropped cornerwise onto a level concrete floor or equivalent hard surface, once on each of its eight corners, falling freely through a vertical distance of 16 inches. At the beginning of each drop, the pilot pack shall be oriented with its center of gravity vertically above the striking corner.

4.6.3.2.2 *Level B.*

4.6.3.2.2.1 *Free-fall drop test.* The pilot pack shall be tested as specified in 4.6.3.2.1.1 for level A, except the drop height shall be 12 inches.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102).

5.1 Pilot pack. The pilot pack is defined as a preproduction pack which will be identical to the pack which the manufacturer or supplier will subsequently produce in fulfillment of the contract. Before a number of junction boxes and covers are packed, a pilot pack shall be made, examined, and subjected to the free-fall drop test specified for the applicable level to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the pilot pack shall not relieve the contractor of his obligation to preserve, package, pack, and mark the junction boxes and covers in conformance to this specification. Any changes or deviations of production packs from the approved pilot pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The junction boxes and covers of like description shall be preserved in accordance with MIL-P-116, method III, in standard commercial quantities, using substantial paperboard cartons as the containers.

5.2.1.1 *Intermediate packaging.* The preserved junction boxes and covers shall be intermediate packaged in quantities not to exceed 40 pounds in boxes conforming to PPP-B-636, type I, class 2, W5c, style optional. The boxes shall be closed and sealed as specified for interior boxes in the appendix to the box specification.

5.2.2 *Level B.* The junction boxes and covers shall be preserved and packaged as specified for level A, except intermediate containers shall be class 1, and closure shall be as specified for domestic, service-grade boxes in the appendix to the box specification.

5.2.3 *Level C.* The junction boxes and covers shall be preserved and packaged in standard commercial quantities in containers that will afford adequate protection against deterioration and damage during shipment from the supply source to the initial destination.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 *Level A.* The junction boxes and covers of like description shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds, and the boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.3.2 *Level B.* The junction boxes and covers of like description shall be packed in close-fitting boxes conforming to PPP-B-621, class 1, style optional; PPP-B-601, domestic type, style optional; or PPP-B-636, type I or II, class 2. The gross weight

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of each wood box shall not exceed 200 pounds, and the boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of fiberboard boxes shall not exceed the weight limitation of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.3.3 *Level C.* The junction boxes and covers shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 **Marking.**

5.4.1 *Civil agencies.* Packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military agencies.* Packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. Nonmetallic junction boxes (commonly known as outlet boxes) and junction-box covers are intended to be used on nonmetallic wiring systems. The boxes are designed to facilitate installation of wires, making wiring connections in electrical systems, and for mounting wiring devices such as switches and receptacles.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and style of junction box required (see 1.2.1.1).

(*c*) Type and style of cover required (see 1.2.2.1).

(*d*) Type of material required (see 3.1).

(*e*) Dimensions of boxes or covers required (see 3.3).

(*f*) Inspection responsibility, if other than specified (4.1).

(*g*) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the junction boxes and covers will be held in covered storage approximately 1 year from date of initial preservation and packaging.

6.3 Definition. In consonance with Cataloging Handbook H6–1, the following definition for junction-box cover shall apply.

6.3.1 *Junction-box cover.* A cover designed to be attached directly to mounting facilities on a junction box. It does not utilize the mounting facilities on wiring devices such as switches and receptacles.

6.4 Sizes of boxes. Sizes of boxes in popular use are indicated in the "Simplified Practice Recommendation," No. R–256–55, issued by the United States Department of Commerce, to which both the purchaser and the supplier are referred. This "Simplified Practice Recommendation" covers metallic junction boxes and is strictly a reference matter for nonmetallic junction boxes.

6.5 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Junction boxes with fittings and covers, porcelain. Carload minimum weight 30,000 pounds.

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Motor:

Junction boxes with fittings and covers, porcelain. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO(ERDL)

Navy—Docks

Air Force—ROM

Review interest: MO, YD, 88

User interest: MC, GL

Copies of this specification may be purchased for cents each.

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